

Praveen Kumar Maddela

Senior Full Stack Developer | React Architect | Lead Software Engineer

Professional Summary

Senior Full Stack Developer | Technology Lead | React Architect | Forward-Deployed Engineer with **8+ years of experience** designing, building, and delivering scalable enterprise applications across Healthcare, SDLC, FinTech, and Product Engineering domains. Proven expertise in **React.js, Next.js, TypeScript, Python, FastAPI, Node.js, REST APIs, PostgreSQL, Cloud Platforms, and AI-powered solutions.**

Experienced in translating business requirements into scalable technical solutions, driving architecture decisions, and leading end-to-end product development from concept through deployment. Strong background in **rapid prototyping, frontend architecture, API design, cloud-native development, system integration, performance optimization, and secure application development.**

Demonstrated success leading engineering initiatives, mentoring developers, conducting architecture reviews, and collaborating with business stakeholders, product teams, designers, and cross-functional engineering groups. Skilled in delivering high-impact solutions through Agile methodologies while maintaining a strong focus on scalability, maintainability, reliability, and user experience.

Skills

AI / Machine Learning

Prompt Engineering, OpenAI API Integration, AI-Assisted Development, LLM-Powered Features, RAG (Exposure), Vector Search, Embeddings, Pinecone, Weaviate, AI Workflow Automation, Agentic Workflow Exposure

Cloud, DevOps & Tools

AWS (EC2, S3), Microsoft Azure, Docker, Kubernetes, GitHub Actions, Azure DevOps, CI/CD Pipelines

Testing & Monitoring: Jest, React Testing Library, Cypress, PyTest, Unit Testing, Integration Testing, Dynatrace, Application Monitoring

Tools & Methodologies: Git, GitHub, GitFlow, VS Code, Agile, Scrum, Code Reviews, Technical Mentoring, Stakeholder Collaboration

Backend Development & Architecture

- Python, FastAPI, Django, Django REST Framework (DRF), Flask, Node.js, Express.js, NestJS, REST APIs, GraphQL, Microservices
- System Design, Service-Oriented Architecture, API Design, Authentication, Authorization, OAuth2, JWT, RBAC, Secure SDLC

Security & Authentication

- **OAuth2**
- **JWT Authentication**
- Role-Based Access Control (RBAC)
- Authentication & Authorization
- XSS Prevention
- CSRF Protection
- SQL Injection Prevention
- Data Encryption

Frontend Architecture & Leadership

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, SCSS, Tailwind CSS, Responsive Design, SPA Architecture, Storybook, Accessibility (WCAG), AG Grid

State Management: Redux, Redux Toolkit, Zustand, Context API, React Hooks, Component-Driven Architecture

Databases, Performance & SEO

PostgreSQL, MySQL, SQL Server, MongoDB, SQLAlchemy, Django ORM, Data Modeling, Query Optimization

Mobile Development, Domains & Certifications

- React Native, Expo; Cross-platform Development (iOS & Android)
- App Deployment & OTA Updates
- **Domain Expertise:** Healthcare, FinTech, E-commerce, PropTech, OTT, Project Management & SDLC Tools
- **Certifications:**
 - AWS Certified Developer – Associate (2025)
 - Microsoft Azure Fundamentals (2024)
 - Advanced React.js & TypeScript (Udemy/Coursera)

Experience

10/2025 – Present

Technology Lead / Full Stack Architect (Forward-Deployed Engineering)

LoopIQ | LoopIQ SDLC Platform

- **Leading architecture, design, and development** of enterprise-grade SDLC and project management platforms using **React.js, Next.js, TypeScript, Python, FastAPI, PostgreSQL, and cloud-native technologies**.
- Partner closely with business stakeholders and product teams to convert evolving requirements into scalable technical solutions and production-ready features.
- Acting as a **Forward-Deployed Engineer**, partnering directly with business stakeholders and product teams to transform evolving business requirements into working prototypes and scalable enterprise solutions.
- Drive rapid feature delivery through iterative build-demo-feedback cycles, enabling faster validation of business requirements and platform enhancements.
- Design reusable frontend frameworks and component libraries using **React Hooks, Redux Toolkit, Zustand, and Context API**.
- Architect and develop secure REST APIs supporting workflow automation, reporting, project governance, and enterprise collaboration.
- Build AI-assisted capabilities and intelligent workflow enhancements through third-party integrations and modern AI development tools.
- Define scalable application architecture, API contracts, data models, and security strategies for enterprise deployments.
- Optimize frontend performance through lazy loading, code splitting, caching strategies, virtualization, and rendering optimizations.
- Design and optimize PostgreSQL and MySQL schemas supporting high-volume transactional workloads.
- Designed and implemented **domain-specific AI assistant capabilities** for SDLC and project management workflows, enabling intelligent task generation, contextual recommendations, and workflow automation.
- Integrated **LLM-powered features** using prompt engineering techniques to improve user productivity and reduce manual effort across project lifecycle activities.
- Evaluated and implemented **vector search and retrieval-based approaches** using Pinecone and AI workflow integrations to support contextual information discovery.
- Collaborated with stakeholders to identify AI use cases and rapidly prototype intelligent solutions through iterative build-demo-feedback cycles.
- Led prototype development initiatives from concept to working demonstrations by collaborating directly with business stakeholders, evaluating technical feasibility, and translating requirements into scalable solution architectures.
- Leveraged modern AI coding assistants and automation tools to accelerate feature development, prototyping, and engineering productivity.
- Participated in architecture discussions around **agentic workflows, tool integration patterns, AI governance, and scalable AI adoption strategies**.
- Implement secure authentication mechanisms using **OAuth2, JWT, RBAC, and enterprise authorization models**.
- Develop backend services and integrations using **Python, FastAPI, Django ORM, and service-oriented architecture patterns**.
- Establish automated testing practices using **Jest, React Testing Library, Cypress, and PyTest**.
- Lead CI/CD implementation using **Docker, Kubernetes, GitHub Actions, and Azure DevOps**.
- Mentor developers through code reviews, architecture guidance, pair programming, and technical leadership initiatives.
- Collaborate with leadership teams to evaluate technical feasibility, identify implementation risks, and recommend scalable solution approaches.
- Drive production reliability improvements through monitoring, troubleshooting, root-cause analysis, and continuous platform optimization.

Key Achievements

- **Improved frontend responsiveness by 35%** through performance optimization initiatives.
- **Reduced deployment effort by 40%** through CI/CD standardization and automation.
- **Accelerated feature delivery cycles** by introducing reusable architecture patterns and shared component frameworks.
- Successfully led architecture decisions supporting enterprise-scale platform growth and multi-team development.

12/2022 – 04/2025

Senior Lead Full Stack Engineer

United Health Group / Optum

- Developed enterprise healthcare platforms using **React.js, TypeScript, Node.js, REST APIs, PostgreSQL, and cloud technologies**.
- Collaborated directly with product owners, architects, business stakeholders, and UX teams to deliver large-scale healthcare solutions.
- Built reusable frontend modules and design patterns improving maintainability and development velocity.
- Developed and integrated secure APIs supporting provider management, healthcare workflows, and enterprise systems.
- Operated in a Forward-Deployed Engineering capacity by collaborating with business stakeholders, product owners, and architects to translate complex healthcare requirements into scalable technical solutions and working application prototypes.
- Participated in architecture reviews and modernization initiatives focused on scalability, reliability, and maintainability.
- Implemented enterprise authentication and authorization frameworks using **JWT, RBAC, and secure identity management practices**.
- Worked on backend integrations involving Python services, workflow orchestration, and API-driven enterprise applications.
- Improved application performance through caching, lazy loading, API optimization, and frontend rendering improvements.
- Contributed to cloud deployment strategies and CI/CD automation using Azure DevOps and containerized environments.
- Created automated testing frameworks and quality engineering practices improving release confidence.
- Supported production systems by analyzing incidents, identifying root causes, and implementing long-term corrective actions.
- Mentored team members, reviewed code, and contributed to engineering best practices across development teams.

Key Achievements

- **Reduced page load times by 30%** through frontend optimization and efficient API consumption strategies.
- **Improved release stability by 25%** through enhanced automated testing and CI/CD improvements.
- Contributed to modernization initiatives impacting multiple healthcare applications across enterprise platforms.

08/2021 – 12/2022

Systems Engineer

TCS (Tata Consultancy Services)

- Developed enterprise web applications using **React.js, Angular, TypeScript, JavaScript, HTML5, and CSS3**.
- Built reusable UI components using **React Hooks, Redux, Context API**, and Angular component architecture.
- Developed responsive and accessible user interfaces supporting enterprise business applications.
- Integrated frontend applications with backend **RESTful APIs** developed using **Node.js** and **Express.js**.

- Collaborated with backend teams to implement end-to-end application features and business workflows.
- Worked with **MongoDB**, SQL Server, and enterprise databases supporting business-critical applications.
- Developed dynamic dashboards, reporting modules, and data visualization components.
- Improved application performance through frontend optimization, code refactoring, and browser performance tuning.
- Participated in testing, debugging, issue resolution, and production support activities.
- Worked with Git, Webpack, Babel, and modern frontend development toolchains.
- Contributed to Agile software development processes including sprint planning, retrospectives, and release activities.
- Conducted peer reviews and collaborated with team members to improve code quality and maintainability.
- Supported deployment activities and coordinated with QA and DevOps teams throughout release cycles.
- Assisted in troubleshooting production issues and resolving application performance bottlenecks.

09/2018 – 08/2021

Full-stack Developer

REDIFCARD TECHNOLOGIES

- Developed full-stack web applications using **React.js**, **Angular**, **Node.js**, **Express.js**, **MongoDB**, HTML5, CSS3, and JavaScript.
- Built and maintained **RESTful APIs** using **Node.js** and **Express.js** supporting customer-facing and internal applications.
- Developed reusable frontend components and interactive user interfaces using React.js and Angular.
- Designed responsive web applications ensuring compatibility across browsers and devices.
- Worked with **MongoDB** and **PostgreSQL** databases for application data management and reporting requirements.
- Participated in database schema design, query optimization, and performance tuning activities.
- Implemented **authentication**, **authorization**, session management, and secure access controls.
- Integrated third-party APIs, payment gateways, notification systems, and external business services.
- Developed administrative dashboards, reporting solutions, and workflow management modules.
- Collaborated with business stakeholders to gather requirements and deliver scalable software solutions.
- Participated in debugging, testing, deployment, and production support activities.
- Worked within Agile teams using Git version control and collaborative development practices.
- Supported application maintenance, feature enhancements, and continuous improvement initiatives.
- Contributed to code reviews, technical documentation, and implementation of software engineering best practices.